

SITUATION REPORT: FLOODS & CONFLICT, CHOCÓ | COLOMBIA DIGITAL TRACE DATA | EXPERIMENTAL



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*Figure 1 — summary displacement map
(to be generated for Chocó)*

Figure 1. Estimated population movement analysis, Chocó, November 2024.

Event Trigger

- Prolonged rainy season intensified by **La Niña** rainfall peaked at **~200 mm/day on 8–9 November**, overflowing the **Atrato, San Juan and Baudó** rivers.
- Warnings limited to **seasonal climate forecasts** (IDEAM) — **no precise localised flood warning** ahead of the event.
- **Compounded by conflict**: an **ELN armed strike** across five municipalities blocked riverine movement, trapping populations in flood-exposed areas and constraining both evacuation and humanitarian access.
- Escalation: Chocó declared public calamity **9 November** a **National Disaster Situation** followed (Decree 1372 of 2024, 13 November).
- Part of a nationwide rainy-season emergency: flooding affected 250,000+ people across at least five departments — Chocó, La Guajira, Antioquia, Santander and Cundinamarca — with Chocó the worst hit.

ESTIMATED DISPLACED · NOV 2024

26,000

people displaced · 27 municipalities

Displacement compounded by an **ELN armed strike** limiting movement for **~20,000** people across five municipalities.

For context: **~215,000** affected in Chocó · **2,100+** homes damaged · IFRC, 2024

Insights

This situation report uses **digital trace data** as an alternative way to measure **flood-driven displacement** in near real-time over the reporting period, applying a GPS-phone approach [1].

Movements are measured as the **deviation from usual mobility** — a design that isolates the change attributable to the floods. Key advantages are **timeliness** and **spatial granularity**.

- **Conflict is central to this event**: an **ELN armed strike** across five municipalities blocked riverine transport for **~20,000 people**, trapping populations in flood-exposed areas, obstructing evacuation and humanitarian access, and **distorting the observed mobility signal**.
- **GPS displacement estimates for Chocó are forthcoming** (Figure 1); remote, low-connectivity riverine terrain makes the signal sparse, and conflict-restricted movement complicates interpretation.
- Displacement fell predominantly on **Indigenous and Afro-Colombian** communities along the Atrato, San Juan and Baudó basins (~70% of Chocó affected).

Situation Report

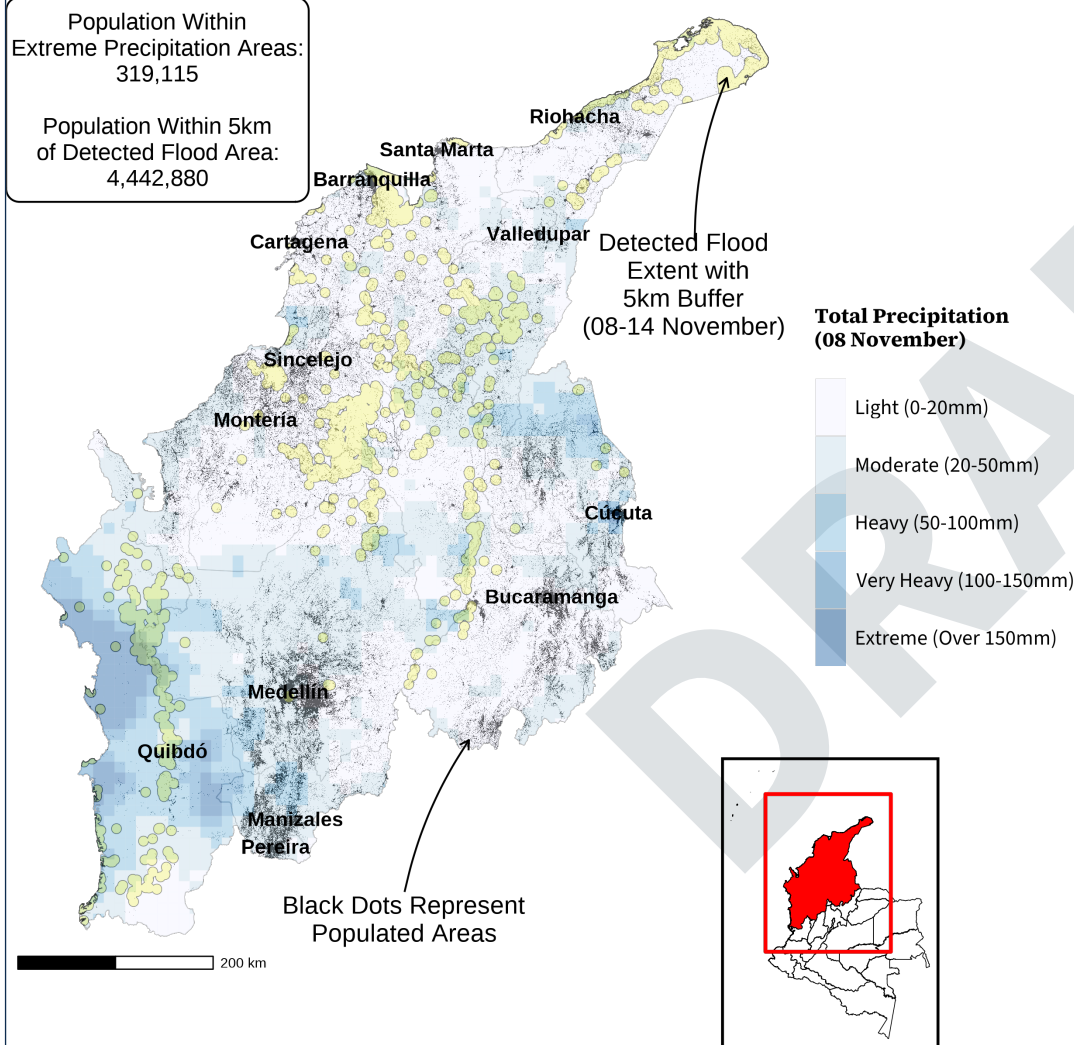
Chocó, Colombia: Floods & Conflict — November 2024

REPORTING PERIOD: 01–21 NOVEMBER 2024

Northern Colombia

Population Within
Extreme Precipitation Areas:
319,115

Population Within 5km
of Detected Flood Area:
4,442,880



Flood context

- **~200 mm/day on 8–9 November** overwhelmed the **Atrato, San Juan and Baudó** basins — Chocó is among the wettest places on Earth — overflowing all three rivers.
- **~70% of Chocó** affected; one of Colombia's poorest departments, with mainly **Indigenous and Afro-Colombian** riverine populations.
- Persistent cloud cover limited satellite detection — Figure 2 **understates the true flood extent**.
- Exposure (to confirm): ~[—] in the high-rainfall area; ~[—] within 5 km of a detected flood.

Warning gap

- Warnings were limited to **seasonal climate forecasts** (IDEAM: 71% probability of La Niña through Q1 2025) — **no precise localised flood warning**.
- IDEAM flagged **700 communities** (>60% of municipalities) at mudslide risk; maximum departmental alerts in **20 municipalities** (Norte de Santander) and **87** (Antioquia).
- **Calamity declarations followed the event** — Chocó on 9 November; National Disaster Situation 10–13 November.

Issuing bodies: IDEAM, UNGRD, departmental DRM offices and IDIGER.

Limitations

- **Low device penetration:** Chocó's remote, dispersed riverine populations have limited smartphone coverage. GPS signal is sparse.
- **Conflict access:** the ELN armed strike restricted river movement, both amplifying harm and potentially distorting observed mobility.
- **Landslide & riverine impacts:** mudslide- and flood-driven displacement is hard to attribute without detailed hazard data.

Patterns are validated against humanitarian reporting (IFRC DREF, 2024). Readers are strongly encouraged to consult the caveats before interpreting these findings.

Figure 2. Rainfall and flood patterns, Chocó, November 2024.

Population: WorldPop, Rainfall: NASA Global Precipitation Measurement (GPM) IMERG Late Run, Flood: NASA Land, Atmosphere Near real-time Capability for Earth observation (LANCE), MODIS satellite

Displacement & conflict

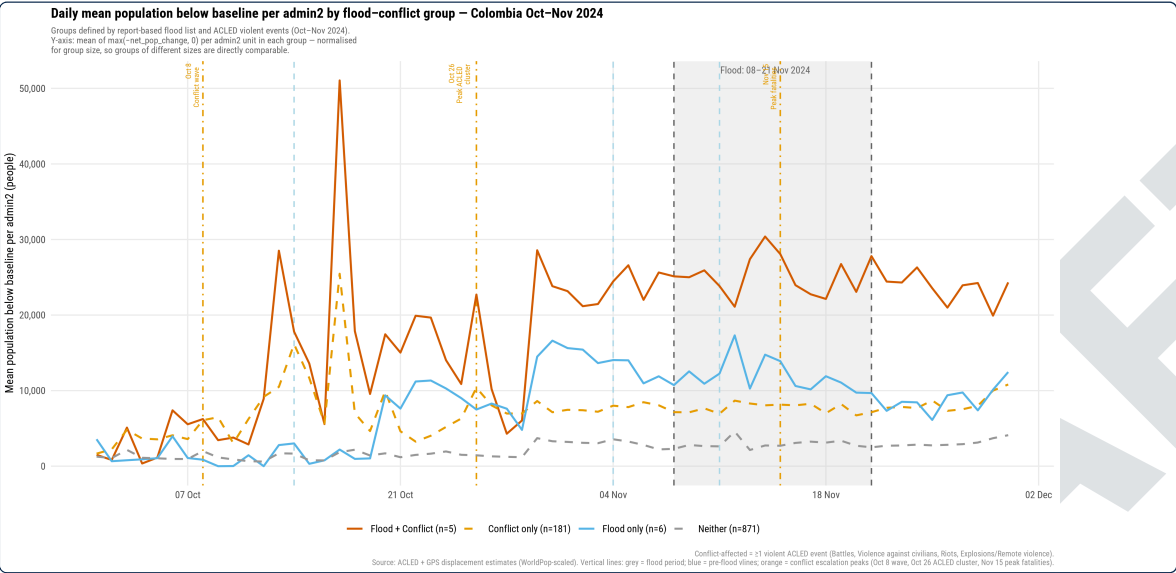


Figure 3. Estimated displacement per unit by joint flood–conflict exposure, Chocó, November 2024. Units are grouped by whether they experienced flooding and/or ACLED-recorded conflict events, isolating the displacement signal where the two hazards overlap.

Data Sources

GPS mobility data. Device-level GPS locations from mobile phones in Chocó, from a commercial vendor aggregating multiple apps to reduce single-source bias. Device counts scaled to population using WorldPop 2020.

Administrative boundaries. OCHA Humanitarian Data Exchange (HDX).

Flood extent. [IFRC | MODIS | GroundSource |.]

Conflict events. Armed Conflict Location & Event Data (ACLED) — geolocated conflict events used to flag conflict-exposed units (Figure 3).

GPS data are not a representative sample; figures exclude populations with limited device access and are indicative of patterns, not precise counts.

Methodology

Step 1. Baseline mobility from pre-flood GPS data (1–7 November 2024).

Step 2. Flooded areas are compared with dry ones to separate flood-driven movement from normal change (8–21 November 2024) (DiD).

Step 3. 1-day displacement threshold; sensitivity checks at 3 and 7 days.

Indicator	Available	Use in this report
GPS / mobile device data	YES	Primary proxy for pop change
MODIS satellite flood data	YES	Flood extent delineation
Copernicus GloFAS	PARTIAL	Intermittent coverage; not used
Cloudflare HTTPS requests	YES	Not used in this report
Google search activity	PARTIAL	Not used in this report
Wikipedia pageviews	PARTIAL	Not used in this report
Night-time lights	PARTIAL	Not used in this report
Meta population estimates	NO	Not used in this report

Table 1. Digital trace data available for Colombia. Sources marked PARTIAL are available but with limited geographic or temporal coverage or resolution.



Bibliography

[1] R. Iradukunda, F. Rowe, and E. Pietrostefani, “Estimating internal displacement in Ukraine from high-frequency GPS mobile phone data,” *Humanities and Social Sciences Communications*, vol. 12, no. 1, p. 1863, 2025.